



Cambridge IGCSE™

MATHEMATICS

Paper 4 (Extended)

MARK SCHEME

Maximum Mark: 100

0580/42

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Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the February/March 2026 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

This document consists of **10** printed pages.

Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptions for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Mathematics-Specific Marking Principles

- 1 Unless a particular method has been specified in the question, full marks may be awarded for any correct method. However, if a calculation is required then no marks will be awarded for a scale drawing.
- 2 Unless specified in the question, non-integer answers may be given as fractions, decimals or in standard form. Ignore superfluous zeros, provided that the degree of accuracy is not affected.
- 3 Allow alternative conventions for notation if used consistently throughout the paper, e.g. commas being used as decimal points.
- 4 Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored (isw).
- 5 Where a candidate has misread a number or sign in the question and used that value consistently throughout, provided that number does not alter the difficulty or the method required, award all marks earned and deduct just 1 A or B mark for the misread.
- 6 Recovery within working is allowed, e.g. a notation error in the working where the following line of working makes the candidate's intent clear.

Annotations guidance for centres

Examiners use a system of annotations as a shorthand for communicating their marking decisions to one another. Examiners are trained during the standardisation process on how and when to use annotations. The purpose of annotations is to inform the standardisation and monitoring processes and guide the supervising examiners when they are checking the work of examiners within their team. The meaning of annotations and how they are used is specific to each component and is understood by all examiners who mark the component.

We publish annotations in our mark schemes to help centres understand the annotations they may see on copies of scripts. Note that there may not be a direct correlation between the number of annotations on a script and the mark awarded. Similarly, the use of an annotation may not be an indication of the quality of the response.

The annotations listed below were available to examiners marking this component in this series.

Annotations

Annotation	Meaning
	More information required
A0	Accuracy mark awarded zero
A1	Accuracy mark awarded one
A2	Accuracy mark awarded two
A3	Accuracy mark awarded three
B0	Independent mark awarded zero
B1	Independent mark awarded one
B2	Independent mark awarded two
B3	Independent mark awarded three
BOD	Benefit of the doubt
C	Communication mark
	Incorrect
FT	Follow through
Highlighter	Highlight a key point in the working
ISW	Ignore subsequent work
M0	Method mark awarded zero
M1	Method mark awarded one
M2	Method mark awarded two
M3	Method mark awarded three

Annotation	Meaning
	Misread
	Omission
Off-page comment	Allows comments to be entered at the bottom of the RM marking window and then displayed when the associated question item is navigated to.
On-page comment	Allows comments to be entered in speech bubbles on the candidate response.
	Premature rounding/approximation
	Special case
	Indicates that work/page has been seen
	Transcription error
	Correct
	Correct answer from incorrect working

MARK SCHEME NOTES

The following notes are intended to aid interpretation of mark schemes in general, but individual mark schemes may include marks awarded for specific reasons outside the scope of these notes.

Types of mark

M Method marks, awarded for a valid method applied to the problem.

A Accuracy mark, awarded for a correct answer or intermediate step correctly obtained. For accuracy marks to be given, the associated Method mark must be earned or implied.

B Mark for a correct result or statement independent of Method marks.

When a part of a question has two or more ‘method’ steps, the M marks are in principle independent unless the scheme specifically says otherwise; and similarly where there are several B marks allocated. The notation ‘dep’ is used to indicate that a particular M or B mark is dependent on an earlier mark in the scheme.

Abbreviations

awrt	answers which round to
cao	correct answer only
dep	dependent
FT	follow through after error
isw	ignore subsequent working
nfw	not from wrong working
oe	or equivalent
rot	rounded or truncated
SC	Special Case
soi	seen or implied

Question	Answer	Marks	Partial Marks
1	7 250 000	1	
2	$x = 106$ $y = 164$	3	B1 for $x = 106$ B2 for $y = 164$ or B1 for angle $ACD = 16$ or M1 for correct method for y e.g. $y = 90 + 74$ or $y = 180 - (90 - 74)$
3	8.7 cao	2	B1 for $\frac{2217}{256}$ or 8.66[0...] or 8.7 seen If 0 scored, SC1 for <i>their</i> more accurate decimal seen correctly rounded to 1 decimal place
4	51 750[.00] final answer	2	B1 for 450 or 89.8... or M1 for $\frac{576}{1.28} \times 115$ oe
5(a)	17 38	2	B1 for 28 08 or 27 68 seen or [0]4 08 or [0]1 20 or 5h 48 m seen or for answer [0]538 [am] or 1738 pm or 17h 38 mins
5(b)	785 or 785.2 to 785.3	2	M1 for $12\,800 \div 16 \frac{18}{60}$ oe
6	0.13 or 13% or $\frac{13}{100}$	2	M1 for $1 - \frac{3}{5} - 27\%$ oe
7(a)	13 or -13 or ± 13	2	M1 for $(-15)^2 - 8 \times 7$ or better
7(b)	$\frac{r^2 - p^2}{8}$ oe final answer	2	M1 for $p^2 - r^2 = -8t$ or better or for correct answer seen then spoilt
8	Pentagon	1	
9(a)	$\frac{4}{3} \times \pi \times 40^3 \div 1000$ oe or $\frac{4}{3} \pi \times 0.4^3 \times 1000$ oe	M2	M1 for $\frac{4}{3} \times \pi \times 40^3$ or $\frac{4}{3} \pi \times 0.4^3$
	268.0[8] to 268.1...	A1	With no errors seen
9(b)(i)	14 [min] 53 [s] cao	3	B2 for 893 to 894 or 14.88 to 14.9 or M2 for $\frac{268000}{300} [\div 60]$ oe or M1 for $268 \div$ figs 3 or figs $268 \div 300$

Question	Answer	Marks	Partial Marks
9(b)(ii)	85.3 or 85.29 to 85.34	3	M2 for $\frac{268000}{\pi \times 50^2 \times 40} [\times 100]$ oe or M1 for $\pi \times 50^2 \times 40$ oe or $\frac{268000}{\pi \times 50^2}$ oe If 0 or M1 scored, SC1 for answer 14.7 or 14.66... to 14.71
10	$-6x + 9y = 117$ oe	B1	
	$2x + 3y = 63$ oe	B1	
	Correct elimination of one variable shown	M1	FT <i>their</i> two linear equations in x and y but not if at least one is of the form $ax + by = 0$
	$x = 6$ $y = 17$	A2	A1 for either If A0 scored, SC1dep for a pair of values that satisfy one of the original equations dep on B1B1 If 0 or B1 scored, SC1 for answers $x = 6$ and $y = 17$
11(a)	$\frac{V - 20}{48[-0]} = 0.25$ or better	M1	
	Correct completion with one more step leading to answer 32	A1	With no errors seen
11(b)	1.52	3	B2 for answer figs 152 or M2 for $\left(\frac{20 + 32}{2} \times 48 + \frac{(65 - 48) \times 32}{2}\right) [\div 1000]$ oe or M1 for $\frac{20 + 32}{2} \times 48$ oe seen isw or $\frac{(65 - 48) \times 32}{2}$ oe seen isw
12	2	3	M1 for eliminating fractions e.g. $3(4x - 3) = 5(7 - 2x)$ or better M1 for reaching $ax = b$ FT <i>their</i> first step

Question	Answer	Marks	Partial Marks
13	$x \geq 3$ oe $y < x - 1$ oe $y \geq -\frac{1}{2}x + 2$ oe	5	B1 for $x \geq 3$ oe B2 for $y < x - 1$ oe or B1 for $y = x - 1$ oe soi B2 for $y \geq -\frac{1}{2}x + 2$ oe or B1 for $y = -\frac{1}{2}x + 2$ oe soi
14	16 cao	3	M2 for $24 \div \sqrt[3]{\frac{1.35}{0.4}}$ oe or M1 for $\sqrt[3]{\frac{1.35}{0.4}}$ oe or $\sqrt[3]{\frac{0.4}{1.35}}$ oe or for $\frac{24^3}{x^3} = \frac{1.35}{0.4}$ oe
15(a)(i)	14, 10, 4	2	B1 for two correct values as answers with maximum 3 values
15(a)(ii)	$\frac{10-x}{2}$ oe final answer	2	M1 for correct first step $x = 10 - 2y$, $y - 10 = -2x$, $\frac{y}{2} = \frac{10}{2} - x$ or better
15(a)(iii)	$8x^2 - 80x + 203$ final answer	3	M1 for $2(10 - 2x)^2 + 3$ oe B1 for $100 - 20x - 20x + 4x^2$ or better or $200 - 40x - 40x + 8x^2$ or better
15(b)	0.5 or $\frac{1}{2}$ oe	1	
16	$a = 80$ $b = 0.955$ or 0.9548 to 0.9549	3	B1 for $a = 80$ B2 for $b = 0.955$ or 0.9548 to 0.9549 or M1 for $20 = \text{their } 80 \times b^{30}$ oe or better soi
17	-0.75 oe nfw	3	B2 for $3x^2 - 5x$ oe or B1 for one correct term $3x^2$ or $-5x$ oe seen
18(a)	27.5	4	M1 for midpoints soi M1 for use of $\sum fm$ with m in correct interval including both boundaries M1dep on 2nd M1 for $\sum fm \div 90$

Question	Answer	Marks	Partial Marks
18(b)	10.4 19.2 1.6	3	B1 for each If 0 scored, SC1 for two of $\frac{13}{10}$, $\frac{36}{15}$, $\frac{6}{30}$ oe seen
18(c)	$\frac{72}{287}$ oe	3	M2 for $\frac{36}{42} \times \frac{6}{41} [\times 2]$ oe or B1 for $\frac{36}{42}$ or $\frac{6}{41}$ or $\frac{6}{42}$ or $\frac{36}{41}$ oe seen or M1 for $\frac{36}{k} \times \frac{6}{k-1} [\times 2]$ or SC1 for answer $\frac{12}{49}$ oe
19(a)(i)	$-q$	1	
19(a)(ii)	$-2p$	1	
19(a)(iii)	$q - p$ final answer	2	M1 for a correct route
19(b)	$2p + 2q$ or $2(p + q)$ final answer	2	B1 for answer $2p + kq$ ($k \neq 0$) or for answer $kp + 2q$ ($k \neq 0$) or for a correct but unsimplified answer
20(a)	113 000 or 113 200 or 113 220 to 113 221	2	M1 for $\frac{1}{2} \times 450 \times 600 \times \sin 57$ oe
20(b)	$\sin[ADC] = \frac{450 \sin 34}{290}$	M2	M1 for $\frac{450}{\sin[ADC]} = \frac{290}{\sin 34}$ oe
	85.80 to 85.81	A2	M1dep on at least M1 for $180 - \text{their } ADC - 34$ oe
20(c)	13.6 or 13.57[0...] nfw	6	B3 for $[BD =] 849$ or 849.2 to $849.3...$ or M2 for $\sqrt{290^2 + 600^2 - 2 \times 290 \times 600 \cos(85.8 + 57)}$ oe or M1 for $290^2 + 600^2 - 2 \times 290 \times 600 \cos(85.8 + 57)$ oe M2dep on M1 for $\tan = \frac{250 - 45}{\text{their } 849}$ oe or M1 for recognition of right-angled triangle with <i>their</i> BD as base and 205 as height

Question	Answer	Marks	Partial Marks
21	11 nfw	3	M2 for $\frac{123.75}{(2 \text{ to } 3) \times (4 \text{ to } 5)}$ or $\frac{123.7 \text{ to } 123.8}{2.5 \times 4.5}$ oe or M1 for 2.5 or 1.5 and 4.5 or 3.5 oe seen or 123.75 or 123.85 oe seen
22	6^{n-2} oe final answer	2	M1 for answer of form $6^{an[+b]}$ oe ($a > 0$) or $\left(\frac{1}{6}\right)^{[b]-an}$ oe ($a > 0$) or for correct answer seen then spoilt
23	39.8 and 219.8 nfw	4	Max 2 marks if angle $A = 39.8$ to 39.81 or 40 or angle $A = 219.8$ to 219.81 or 220 shown B3 for 140.2 and 320.2 as answers or shown in working or M3 for answer 39.8 with 140.2 or 320.2 shown in working or as answer or for answer 219.8 with 140.2 shown in working or as answer or B2 for 140.2 or 320.2 or M1 for $\tan A = \frac{-2-0.5}{3}$ oe If M1 or 0 scored, SC1 for two answers with a difference of 180° that lie in the range 0 to 360